LATENTLAB

Unlocking Underused Research Equipment, Scientific Labs & Spaces

Problem Statement: Unlock Underused Equipment and Spaces	**Team Name:** Analytical Eye
Sole Member & Lead: Pankhuri Govila	**Format:** 48-Hour Continuous Build Hackathon
Live App URL: https://latent-lab-unlock-underused-equipme.vercel.app/	**GitHub Repo:** github.com/pgovila/Latent-Lab---Unlock-Underused-Equipment-and-Spaces

1. Understanding of the Problem & Intended Users

Institutional Problem Context: Universities, technical colleges, research institutions, and community makerspaces house tens of millions of dollars in highly specialized apparatus and spaces. However, research reveals that instruments such as Next-Gen Genomic Sequencers (Illumina NovaSeq), industrial CO2 laser cutters, multi-GPU AI compute clusters, cleanrooms, and cine-optics packages operate at less than **25% to 35% capacity** outside core departmental hours. Concurrently, undergraduate researchers, cross-departmental teams, and early-stage innovators are unable to access this equipment due to prohibitive commercial rental fees or siloed departmental ownership.

The Trust, Safety & Operational Gap: Generic calendar or booking tools (e.g., Google Calendar, Calendly) fail in institutional research settings because they lack: (1) Mandatory safety qualification and training badge verification (e.g., BSL-2 biosafety, laser safety certification, cleanroom protocols); (2) Verified before-and-after digital custody records (identifying who damaged delicate apparatus); and (3) Anti-monopolization controls (preventing single research groups from hoarding shared compute nodes or laser beds).

Target Users & Key Personas:

- **Student & Faculty Researchers:** Need friction-free discovery of available lab apparatus across campus, transparent qualification checks, and fair allocation without out-of-pocket billing.
- **Resource Owners (PIs & Lab Managers):** Need full oversight on who operates their hardware, transparent logged approval/rejection workflows, and automated assurance that operators hold valid safety credentials.
- **Campus Administrators & Research Integrity Officers:** Need an immutable, auditable transaction ledger for dispute arbitration, penalty enforcement, and equipment utilization metrics.

2. Proposed Solution & Complete Product Workflow

LatentLab provides an end-to-end trusted resource-sharing ecosystem engineered around a **Hybrid Sponsored Credits Model** (100 monthly credits per researcher) and a verifiable digital chain of custody. The entire 10-step lifecycle operates sequentially:

Step	Milestone	Implementation & Operational Flow
1	**Owner Listing Dashboard**	Owners list equipment, set credit costs, availability slots, and mandate training badges.
2	**Discovery & Match**	Researchers filter by category/location with real-time badge checks ('Verified Eligible' vs 'Missing Badge').
3	**Booking Engine**	Validates available credits and enforces anti-monopolization quota (maximum 3 concurrent active bookings).
4	**Auto-Verification Layer**	Evaluates safety rules. Runs AI Conflict Resolution; places congested slots in a transparent queue.
5	**Owner Review Portal**	Custodians approve or reject with logged reason notes. Rejections automatically restore user credits.
6	**State Transition Banner**	Real-time visual banner alerts researchers of approvals, queue promotions, or status changes.
7	**Digital Check-In**	Pre-unlock inspection: logs baseline 1-5★ rating, condition notes, photo URL, and mandatory PPE compliance.

8	**Digital Return / Relock**	Final inspection rating, sanitization notes, and automated late penalty calculation (5 credits / 15 min).
9	**Dispute & Incident Hub**	Reports damage/delays. Vercel AI SDK parses narrative descriptions into Low/Med/High severity triage.
10	**Auditable Ledger Timeline**	Immutable, chronological visual audit trail logging actor, timestamp, action code, and JSON metadata.

3. Features Completed vs. Anything Left Unfinished

Fully Completed Features (100% Operational):

- **Complete 10-Step Lifecycle:** From resource listing, qualification match, booking, owner review, digital check-in/out, dispute handling, to the auditable ledger.
- **Real-Time Persona Switcher:** Instant switching between Pankhuri Govila (Researcher), Dr. Aris Thorne (Lab Owner), Elena Rostova (Makerspace Mgr), and Campus Admin to evaluate both sides of the lifecycle effortlessly.
- **Conflict Resolution AI:** Detects scheduling overlaps and synthesizes optimal alternative slots (Earliest Available with buffer, or Next-Day identical slot).
- **Incident Summarizer AI:** Natural language severity triage (Low/Med/High) and repair recommendations.
- **Cross-Campus Federated RAG Pipeline:** Multi-university semantic vector retrieval connecting Stanford Bio-X, MIT.nano, UC Berkeley Marvell, and the National AI Consortium with reciprocal safety credential reconciliation.
- **Anti-Monopolization Safeguards:** Strict 3 active bookings cap per researcher and deterministic waitlist positioning.
- **Production PostgreSQL Schema:** Complete RLS policies, tables, indexes, and constraints in `schema.sql`.
- **Hybrid Persistence Engine:** Resilient in-memory seed synchronization enabling judges to evaluate the app immediately without requiring third-party cloud database configurations.

Deliberate Boundaries & Known Limitations (Left for Production Scaling):

- **Physical IoT Relay Hardware:** Digital unlock transitions are simulated in software state rather than firing physical GPIO magnetic door strikes.
- **University SSO Integration:** Active personas use pre-verified campus directory profiles rather than SAML 2.0 / Shibboleth identity providers.

4. Technology Stack & Implementation Details

- **Frontend & App Router:** Next.js 16 (React 19, Turbopack, TypeScript) utilizing a monolithic component architecture to prevent assembly errors.
- **Styling & Aesthetics:** Tailwind CSS, Lucide React, and glassmorphism design tokens providing a modern, institutional dark mode aesthetic.
- **Database & Security:** Supabase PostgreSQL with granular Row Level Security (RLS) policies protecting `profiles`, `resources`, `bookings`, `condition_logs`, and `activity_ledger`.
- **AI & Federated RAG Engine:** Vercel AI SDK integration (`/api/ai` route) and multi-campus semantic vector retrieval (`queryFederatedRAG`) over disparate university operating manuals and safety SOPs.
- **Local Environment & Testing:** Isolated Python 3.12 virtual environment (`.venv`) with automated runner and verification suite (`run_app.py`).

5. Testing Notes Covering Main Product Journeys

The system has undergone end-to-end regression testing across the primary user journeys:

- **Journey A (Researcher Discovery & Reservation):** Filtered resources by category, validated badge eligibility check, verified AI conflict resolution, confirmed credit deduction (-35 credits for NovaSeq), and ensured max-3 active booking enforcement.
- **Journey B (Owner Review & Decision):** Switched persona to Dr. Thorne, reviewed pending booking request, entered decision note, approved request, verified state transition banner update, and tested credit refund upon rejection.
- **Journey C (Digital Custody Check-In & Check-Out):** Switched to approved booking, submitted 5★ rating with baseline notes and PPE certification, verified resource unlock state. Completed check-out, confirmed automated overdue penalty calculation

formula, and verified resource availability restoration.

- **Journey D (Dispute & Incident Triage):** Submitted equipment damage description, verified AI categorization as [HIGH SEVERITY], and confirmed automatic escalation to the admin ledger.
- **Journey E (Automated Test Suite):** Executed `python run_app.py --test` in Python 3.12 virtual environment, confirming HTTP 200 health check and valid JSON responses from both AI endpoints.
- **Journey F (Cross-Campus Federated RAG):** Executed natural language research queries (Cryo-TEM, direct-write lithography, deep silicon etching), verified semantic relevance scoring, checked reciprocal credential verification, and completed cross-campus passport reservation.

6. Future Scope & Possible Improvements

- **Physical IoT Smart-Locks & Smart-Plugs:** Integrate ESP32/Raspberry Pi microcontrollers to electronically energize high-voltage equipment only upon completed Milestone 7 Check-In.
- **Institutional SSO & LMS Integration:** Connect to Canvas/Blackboard to automatically sync safety training quiz completion into verified profile badges.
- **Cross-Institutional Consortium Sharing:** Expand credit federation across regional university clusters (e.g., sharing underutilized cleanrooms between neighboring research universities).
- **Predictive Maintenance AI:** Analyze condition log rating degradation trends over time to schedule preventive servicing before catastrophic failure occurs.

7. Complete Team & Participant Details

Role / Parameter	Official Hackathon Details
Team Name	Analytical Eye
Team Composition	Solo Participant (1 Member)
Full Name	Pankhuri Govila
Email	pankhgovila@gmail.com
Live App URL	https://latent-lab-unlock-underused-equipme.vercel.app/
Institution / Track	We Code Coders · 48-Hour Open Innovation Hackathon
Problem Statement	Unlock Underused Equipment and Spaces
Repository URL	https://github.com/pgovila/Latent-Lab---Unlock-Underused-Equipment-and-Spaces

8. Member's Role & Personal Contribution

As the sole developer and team lead of **Analytical Eye**, **Pankhuri Govila** conceived, designed, developed, and tested 100% of the project:

- **System Ideation & Architecture (100%):** Formulated the core university lab use case, the 100-credit sponsored economic model, the 3-booking anti-monopolization quota, and the 10-step digital custody workflow.
- **Full-Stack Development (100%):** Built the Next.js 16 App Router monolithic application, designed responsive glassmorphism UI components, implemented client-side state mutation engines, and structured the role-switching architecture.
- **Database & Security Engineering (100%):** Authored the production PostgreSQL schema (`schema.sql`), defined relational constraints, and established Row Level Security (RLS) policies for Supabase.
- **AI Engineering (100%):** Integrated Vercel AI SDK modules for Conflict Resolution and NLP Incident Summarization with heuristic fallback engines.
- **Testing & Documentation (100%):** Configured Python 3.12 runner and verification suite (`run_app.py`), authored judges' evaluation walkthroughs, and assembled the official project submission documentation.